

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1299

Timken 100 kV
November 15, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1299. The Interconnection Customer proposes a 58.2 MW battery energy storage facility interconnecting at the Timken 100 kV substation in the Ohio area. The request was received on September 27, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 463 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 24 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$81,164,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Madisonburg-Vitulli 100kV Line 1	Madisonburg-Vitulli 100kV #2	181.8	152	6.7
Madisonburg-Vitulli 100kV #2	Madisonburg-Vitulli 100kV	181.8	152	6.7
Vitulli-Ochiltree 100kV	Trego-Beaver 500kV	159.2	135	7.9
Smuckers-Ochiltree 100kV	Trego-Beaver 500kV line	159.1	135	7.9
Grassland-Smuckers 100kV	Trego-Beaver 500kV	149.9	127	7.9
Franchise Tax-Northern States Flambeau 100kV line	Franchise Tax-Northern States Flambeau 100kV Line 2	147.1	156	9.6
Dalton-Grassland 100kV	Beaver-Trego 500kV	145.9	123	8.7
Dublin-Dalton 100kV line	Trego Substation-Beaver 500kV	144.6	114	8.7
Northern States Flambeau Substation-Franchise Tax 100kV Ckt 2	Franchise Tax-Northern States Flambeau 100kV	143.9	160	9.6
Los Vaqueros-Colleyville 100kV Line	W Coshocton-West Astoria 100kV	134.8	138	12.2
Franchise Tax-Colleyville 100kV	Colleyville-Franchise Tax 100kV Ckt 2	129.2	149	11.3
Franchise Tax-Colleyville 100kV Cir. 2	Colleyville Substation-Franchise Tax 100kV Ckt 1	129.2	149	11.3
Rosedale-Billings Eastside 100kV	Ridgely-Sutliff 100kV L	126.4	159	16.1
Leetonia-J M Huber 230kV	El Paso-Ravenna 500kV	122.3	534	6.2
Los Vaqueros-Tewksbury No. 22A 100kV	Trail-Colleyville 100kV	117.5	137	17.1
West Astoria-West Coshocton 100kV	Los Vaqueros-Colleyville 100kV	117.1	143	6.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
South Coshocton-West Astoria 100kV	Los Vaqueros-Colleyville 100kV	109.9	140	6.6
Mcdonnell-Rosedale 100kV	Lakemore-Simmons 100kV Line	109.3	118	5.6
Benson Power Biomass Plant-Tewksbury No. 22A 100kV	Trail-Colleyville 100kV	105.0	132	17.1
Sugar Island-Rosedale 100kV	Ridgely-Rosedale 100kV	104.4	121	10.8
Walter Bouldin Dam-S Coshocton 100kV	Los Vaqueros-Colleyville 100kV	103.1	137	6.6
Colleyville-Trail 100kV	Los Vaqueros-Colleyville 100kV	102.6	130	6.2
Benson Power Biomass Plant Substation-Reedurban 100kV	Trail-Colleyville 100kV	101.1	129	17.1
Burwell 500/345kV Autotransformer	Allegheny Energy Units 3 4 & 5 345 kV/1 kV XFMR	100.0	1,191	6.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Madisonburg-Vitulli 100kV Line 1	Rebuild line with higher-capacity conductor	\$7,428,000	17% (\$1,277,000)
Madisonburg-Vitulli 100kV Cir. 2	Rebuild line with higher-capacity conductor	\$9,105,000	17% (\$1,565,000)
Vitulli-Ochiltree 100kV	Rebuild line with higher-capacity conductor	\$443,000	100% (\$443,000)
Ochiltree-Smuckers 100kV	Rebuild line with higher-capacity conductor	\$3,045,000	100% (\$3,045,000)
Smuckers-Grassland 100kV L	Rebuild line with higher-capacity conductor	\$2,717,000	100% (\$2,717,000)
Franchise Tax-Northern States Flambeau 100kV line	Rebuild line with higher-capacity conductor	\$7,009,000	100% (\$7,009,000)
Dalton-Grassland 100kV	Rebuild line with higher-capacity conductor	\$7,651,000	100% (\$7,651,000)
Dublin-Dalton 100kV line	Rebuild line with higher-capacity conductor	\$2,945,000	100% (\$2,945,000)
Northern States Flambeau Substation-Franchise Tax 100kV Ckt 2	Rebuild line with higher-capacity conductor	\$9,213,000	100% (\$9,213,000)
Los Vaqueros-Colleyville 100kV L	Rebuild line with higher-capacity conductor	\$3,441,000	100% (\$3,441,000)
Franchise Tax-Colleyville 100kV Line 1	Rebuild line with higher-capacity conductor	\$13,145,000	100% (\$13,145,000)
Franchise Tax-Colleyville 100kV Circuit 2	Rebuild line with higher-capacity conductor	\$10,380,000	100% (\$10,380,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rosedale-Billings Eastside 100kV	Rebuild line with higher-capacity conductor	\$2,931,000	6% (\$163,000)
Leetonia-J M Huber 230kV	Rebuild line with higher-capacity conductor	\$19,063,000	5% (\$975,000)
Los Vaqueros-Tewksbury No. 22A 100kV L	Rebuild line with higher-capacity conductor	\$4,086,000	100% (\$4,086,000)
West Astoria-West Coshocton 100kV	Rebuild line with higher-capacity conductor	\$4,095,000	100% (\$4,095,000)
South Coshocton-W Astoria 100kV line	Reconductor line	\$578,000	100% (\$578,000)
Rosedale-Mcdonnell 100kV line	Reconductor line	\$2,324,000	3% (\$80,000)
Benson Power Biomass Plant-Tewksbury No. 22A 100kV	Reconductor line	\$1,456,000	100% (\$1,456,000)
Sugar Island-Rosedale 100kV	Reconductor line	\$1,882,000	24% (\$444,000)
Walter Bouldin Dam-S Coshocton 100kV	Reconductor line	\$2,550,000	100% (\$2,550,000)
Colleyville-Trail 100kV	Reconductor line	\$2,581,000	100% (\$2,581,000)
Benson Power Biomass Plant Substation-Reedurban 100kV	Reconductor line	\$1,214,000	100% (\$1,214,000)
Burwell 500-345 kV Autotransformer	Replace transformer with higher-rated unit	\$15,128,000	1% (\$111,000)
	Total allocated to project		\$81,164,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.